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VPC Traffic Flow and Security



yakubu abdullahi yusuf

Security group (sg-0e137d8c08560fd13 | NextWork Security Group) was created successfully

Details

sg-0e137d8c08560fd13 - NextWork Security Group

Actions

Details

Security group name NextWork Security Group	Security group ID sg-0e137d8c08560fd13	Description A Security Group for the NextWork VPC	VPC ID vpc-016d9b867bab64aff
Owner 176971016201	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules

Outbound rules

Sharing

VPC associations

Tags



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources securely. It is useful because it gives you control over networking and security, allowing you to decide how resources communicate with each other and with the internet.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to set up a private network, create subnets, configure an internet gateway and route tables, and apply security groups and network ACLs to control and secure traffic to my cloud resources.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how much control and flexibility one has over networking and security using Amazon VPC.



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This project took me...

This project took me thirty to forty minutes to complete.



Route tables

Route tables are set of rules that decide how data moves between subnets, the internet, and other networks.

Route tables are needed to make a subnet public because they define a route that allows traffic from the subnet to reach the internet through an Internet Gateway.

The screenshot shows the 'Edit routes' interface in the AWS Management Console. The breadcrumb navigation at the top indicates the path: VPC > Route tables > rtb-0c8dc471888b33880 > Edit routes. The main section is titled 'Edit routes' and contains a table with columns: Destination, Target, Status, Propagated, and Route Origin. There are two rows of routes. The first row has a Destination of '10.0.0.0/16', a Target of 'local', a Status of 'Active' (indicated by a green checkmark), and a Route Origin of 'CreateRouteTable'. The second row has a Destination of '0.0.0.0/0', a Target of 'Internet Gateway', a Status of '-' (inactive), and a Route Origin of 'CreateRoute'. Below the table, there is an 'Add route' button. At the bottom right, there are 'Cancel', 'Preview', and 'Save changes' buttons.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes



Route destination and target

Routes are defined by their destination and target, which mean the destination says where the traffic wants to go, and the target says how it gets there.

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and a target of igw-065c0f068bae50abf.

VPC > Route tables > rtb-0c8dc471888b53880 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes



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Security groups

Security groups decide what traffic is allowed in and out of your resources.

Inbound vs Outbound rules

An inbound rule defines what incoming traffic is allowed to reach your resources, such as EC2 instance. It specifies who can connect, using which protocol, and on which port. My security group allows inbound HTTP traffic from anywhere on the internet so users can access the web application.

Outbound rules are rules that defines what traffic is allowed to leave a resource such as an EC2 instance. It controls where the instance can send data. By default, my security group's outbound rule is to any destination (0.0.0.0/0)



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Network ACLs

Network ACLs (Access Control Lists) are an extra layer of security that control inbound and outbound traffic at the "subnet level" in a VPC.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security groups are at the resource level while network ACL are at the subnet level.



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffics in and out of the subnet.

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all traffic.

Details info

Network ACL ID

acl-011053acff94c0be8

Associated with

subnet-0413f5192d99a28ec / Public 1 - subnet

Default

No

VPC ID

vpc-016d9b867bab64aff / NextWork VPC

Owner

176971016201

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (2)

Edit inbound rules

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



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Tracking VPC Resources

I created additional VPC, an internet gateway and security groups. Instead of my usual region, I used AWS CLI/ cloud shell to create the resources. Teams would use multiple regions to improve availability, reduce latency for users in different locations, and provide disaster recovery in case one region fails.

EC2 Global View is a tool where you can find all your EC2 resources across different AWS regions in one place. I could even narrow down my search by region, instance state, or tags. Without EC2 Global View, you'd have to manually switch between regions to check your instances one by one.

Now that I've learnt about EC2 Global View, I'd use it again to monitor and manage my EC2 instances across multiple regions without switching regions manually.



AWS Global View		Resource totals will be inaccurate until complete			
Region Explorer		Compute	Storage	Networking	Security
Global search		0	0	386	47
Regions and Zones		Show all resource summary			
Settings		Enabled regions	Instances	VPCs	Subnets
		19 regions	0 in 0 regions	21 in 19 regions	62 in 19 regions
		Security groups	Volumes	Auto scaling groups	Route tables
		25 in 19 regions	0 in 0 regions	0 in 0 regions	21 in 19 regions
		VPC endpoints	NAT gateways	Egress only internet gateways	Internet gateways
		0 in 0 regions	0 in 0 regions	0 in 0 regions	21 in 19 regions
		DHCP option sets	Elastic IPs	Endpoint services	Managed prefix lists
		19 in 19 regions	1 in 1 regions	0 in 0 regions	241 in 19 regions
		Network ACLs	Network interfaces	VPC peering connections	Capacity Reservations
		22 in 19 regions	0 in 0 regions	0 in 0 regions	0 in 0 regions
		S3 buckets	RDS clusters	RDS DB instances	Outposts
		0 in 0 regions	0 in 0 regions	0 in 0 regions	0 in 0 regions
To discover more resources, visit AWS Resource Explorer					



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